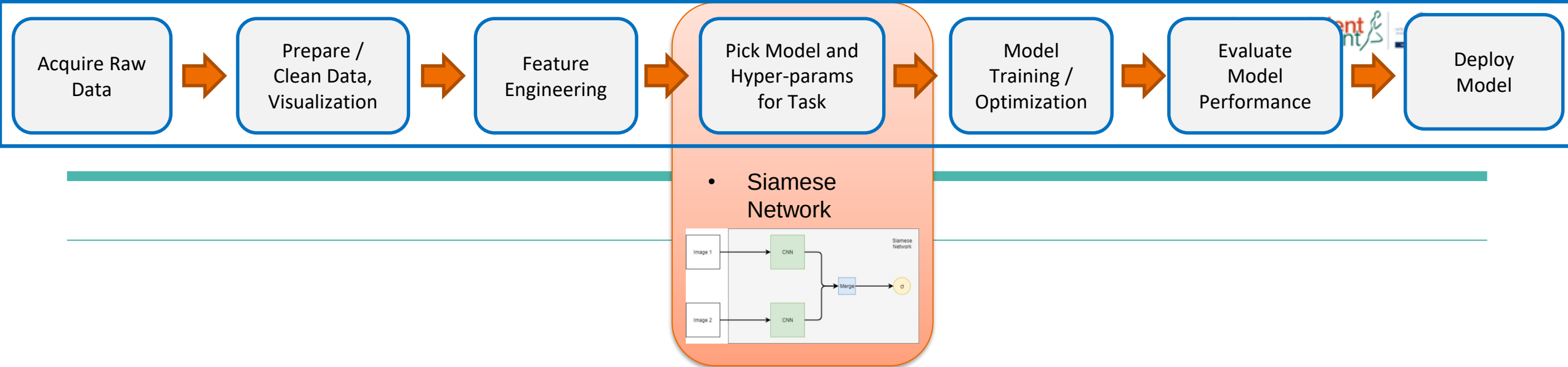


# Learning Settings



## Siamese and Loss Functions

# Recognition Vs Verification

- Recognition: eg. K–Class recognition
  - Given a sample, decide which one of the K it is
- Verification: Given a sample and a class ID,
  - Say “YES” or “NO”
- What makes these two different?
  - Popular in many situations (Eg. Biometrics)
    - Face, Fingerprint
  - An Intra-class variation could be higher than inter-class
    - People change appearance over time.

# Verification: Test/verification time

- Often done with nearest neighbors
  - Too much of computing?
- Find “K” nearest neighbors Find the distance to the samples from the class under question
  - Find Mean distance
  - Find Min Distance
- Variable Threshold: Subject specific
  - Some people have high variability in “signature”

# Verification: Enrolment/Training

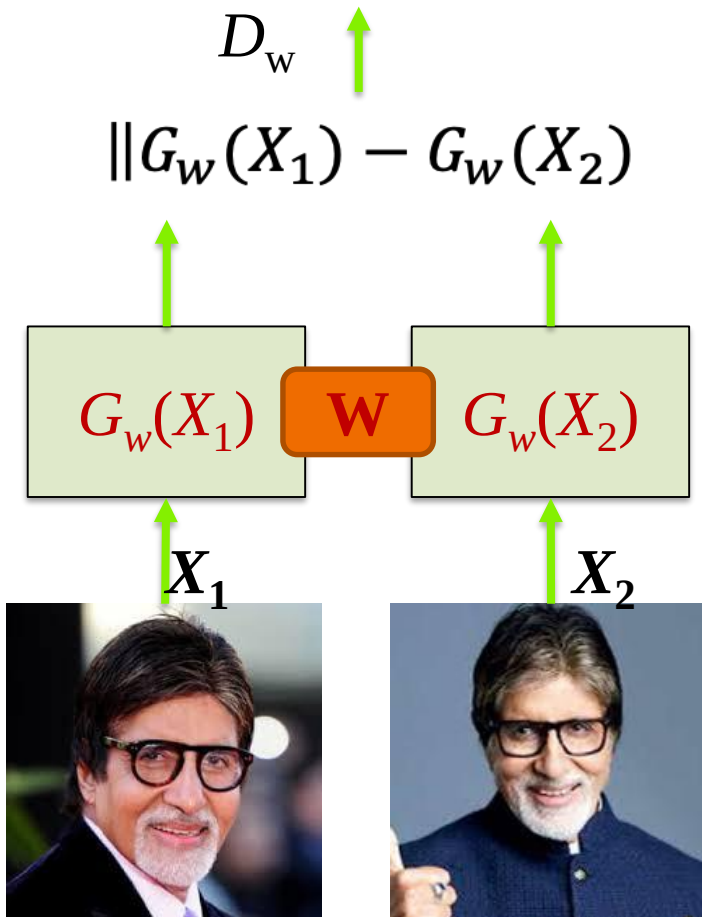
- Usually only small number of samples to add
  - One shot learning (single sample setting)
- Find a feature space that suits the local “geometry”
  - Metric learning
  - Aka finding an appropriate distance function

# Points to Notice

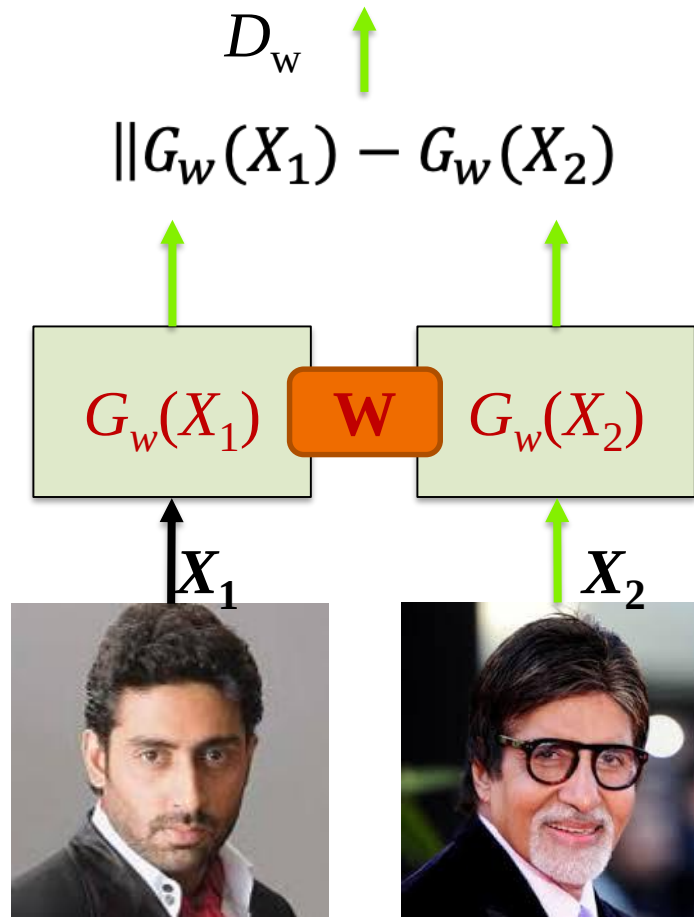
- Features that can capture “Fine Variation”
- Classification Vs Verification
- Learning with a “different style” of Supervision
- Contrastive Learning and Self Supervised Learning

# Siamese Architecture/Loss

Make this smaller



Make this larger



- Only pair-wise Labels
- Similarity Metric:  
 $D_w(X_1, X_2)$
- Have shared weights
- Training in batches

# Historic

- **Input:** A pair of input signatures.
- **Output (Target):** A label, **0** for similar, **1** else.

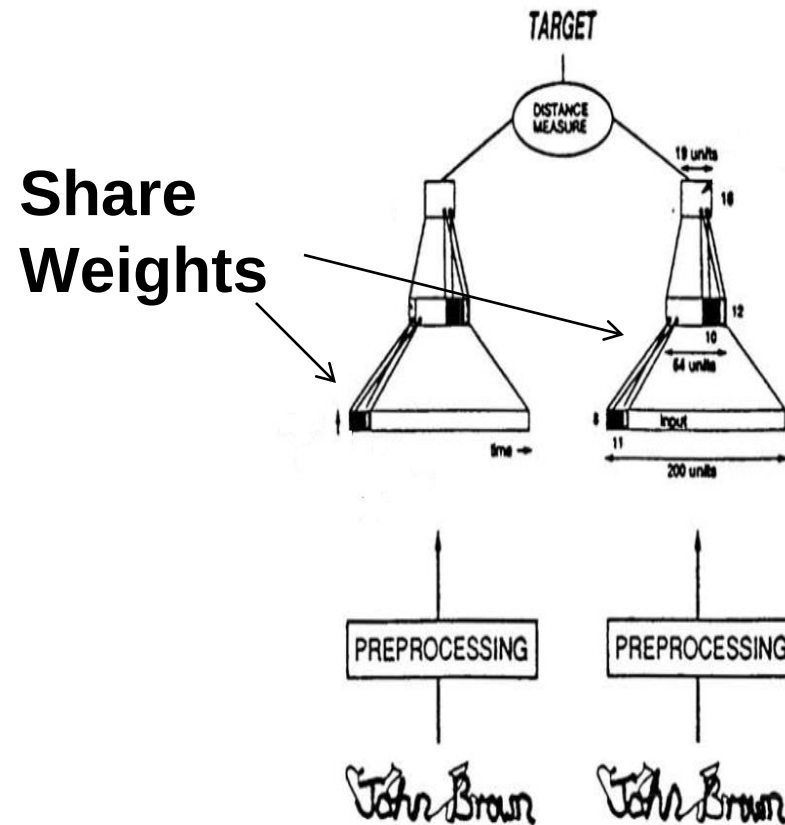
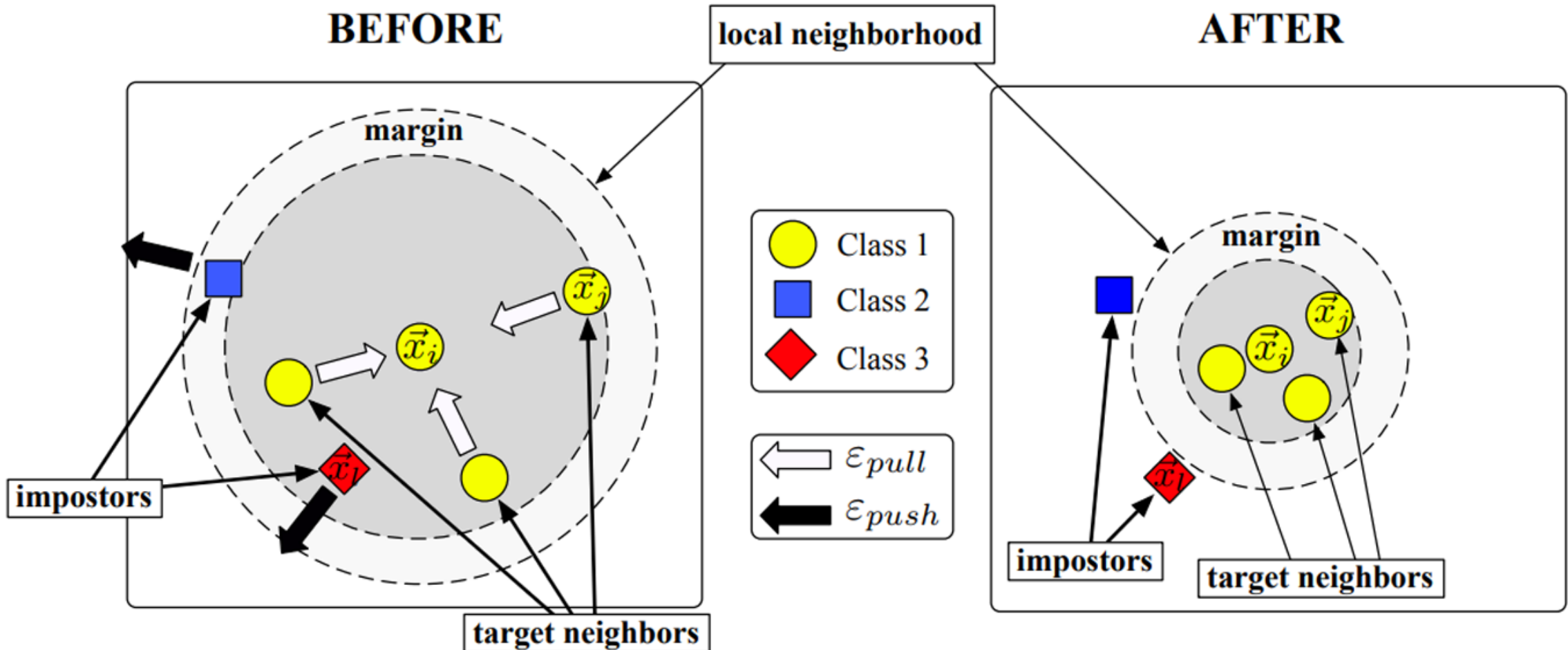


Image Source:  
Google

Bromley, J., Bentz, J.W., Bottou, L., Guyon, I., LeCun, Y., Moore, C., Säckinger, E. and Shah, R., 1993. Signature Verification Using A "Siamese" Time Delay Neural Network. *IJPRAI*, 7(4), pp.669-688.

# Historic: Large Margin Nearest Neighbour(LMNN)



Intuition in Nearest Neighbor Setting

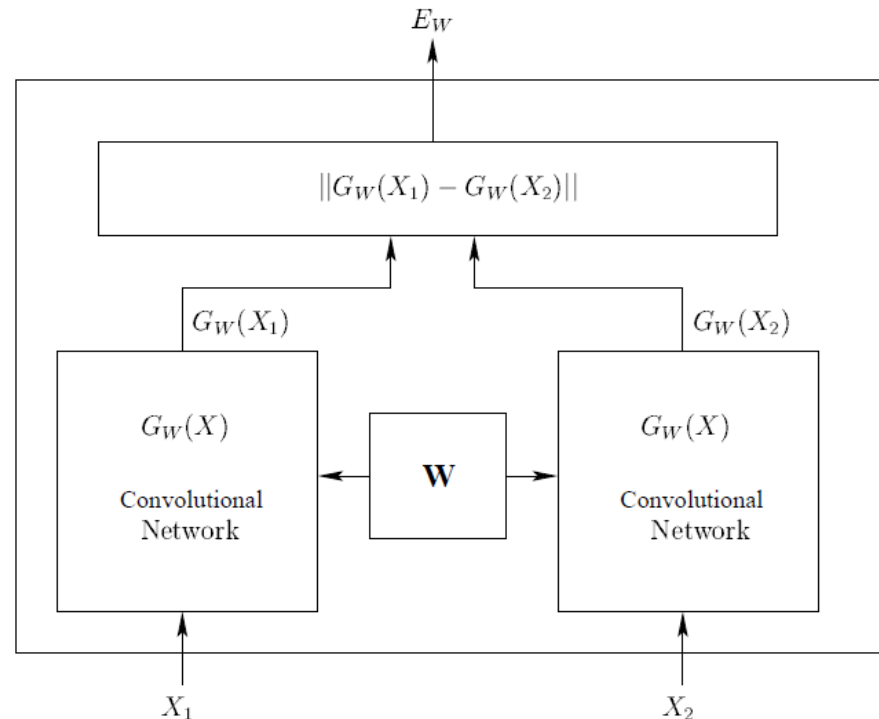
# Blank Slide



# Siamese Architecture

- Given a family of functions  $G_W(X)$  parameterized by  $W$ , find  $W$  such that the similarity metric  $D_W(X_1, X_2)$  is small for similar pairs and large for dissimilar pairs:-

$$D_W(X_1, X_2) = \|G_W(X_1) - G_W(X_2)\|$$

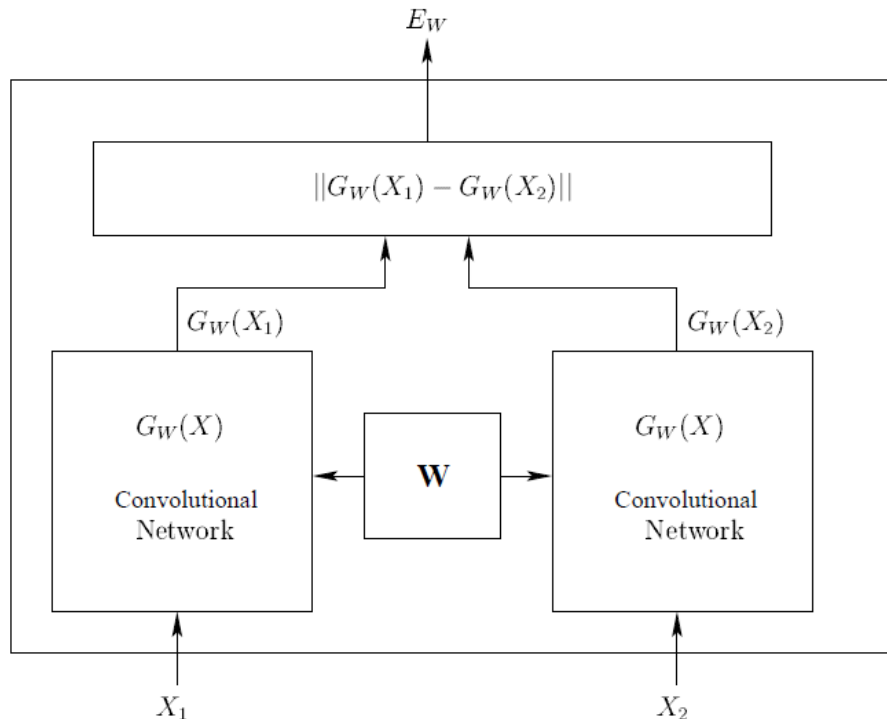


# Siamese Architecture

- Given a family of functions  $G_W(X)$  parameterized by  $W$ , find  $W$  such that the similarity metric  $D_W(X_1, X_2)$  is small for similar pairs and large for dissimilar pairs:-

$$D_W(X_1, X_2) = \|G_W(X_1) - G_W(X_2)\|$$

$Y=0$  for “similar” and  $Y=1$  for “dissimilar” pairs.



## Loss function

$$\mathcal{L}(W) = \sum_{i=1}^p L(W, (Y, \vec{X_1}, \vec{X_2})^i)$$

## Loss function for similar pairs

$$L(W, (Y, \vec{X_1}, \vec{X_2})^i) = (1 - Y)L_s(D_W^i) + YL_D(D_W^j)$$

Loss function for dissimilar pairs

# Blank Slide

# Contrastive Loss function

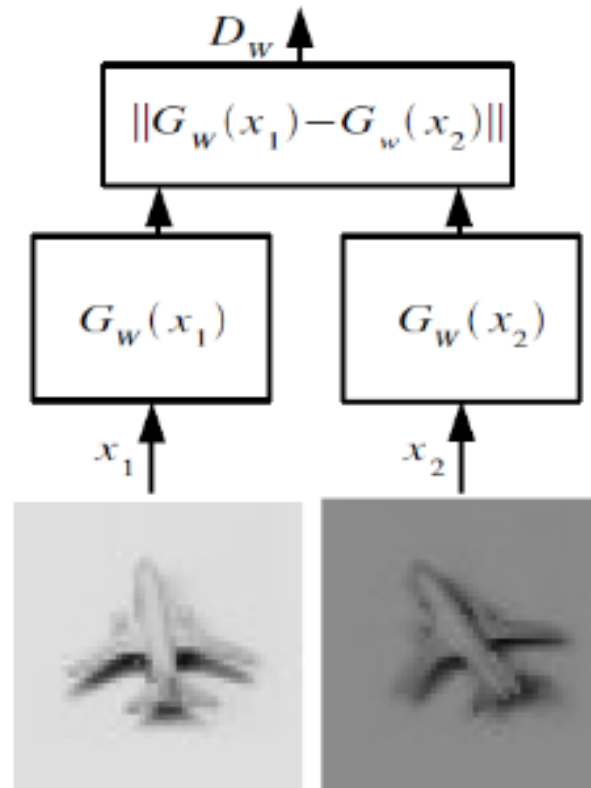
- Given  $D_W(X_1, X_2)$  is the distance between the pair of samples, the contrastive loss function is defined as follows: (Y=0 for Similar and Y=1 for dissimilar pairs)

$$(1 - Y) \frac{1}{2} (D_W)^2 + (Y) \frac{1}{2} \{\max(0, m - D_W)\}^2$$

# Blank Slide

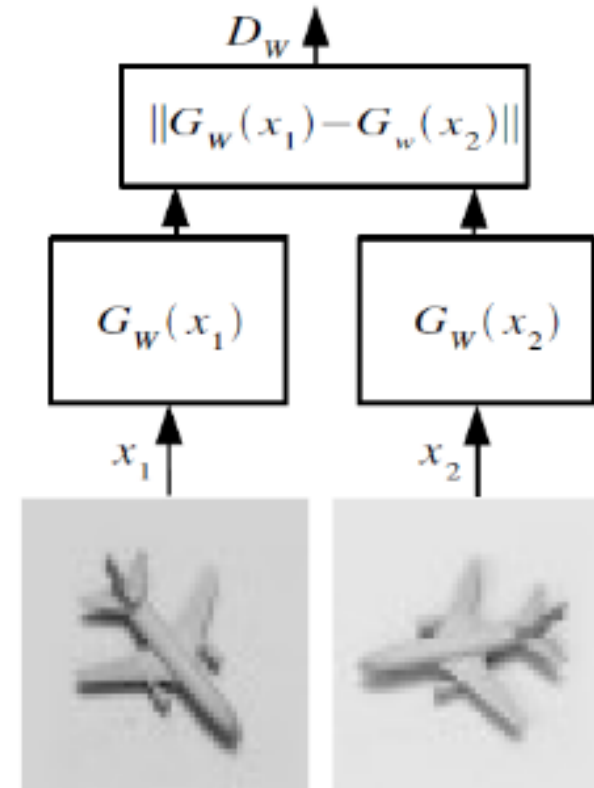
# Siamese: Loss

Make this small



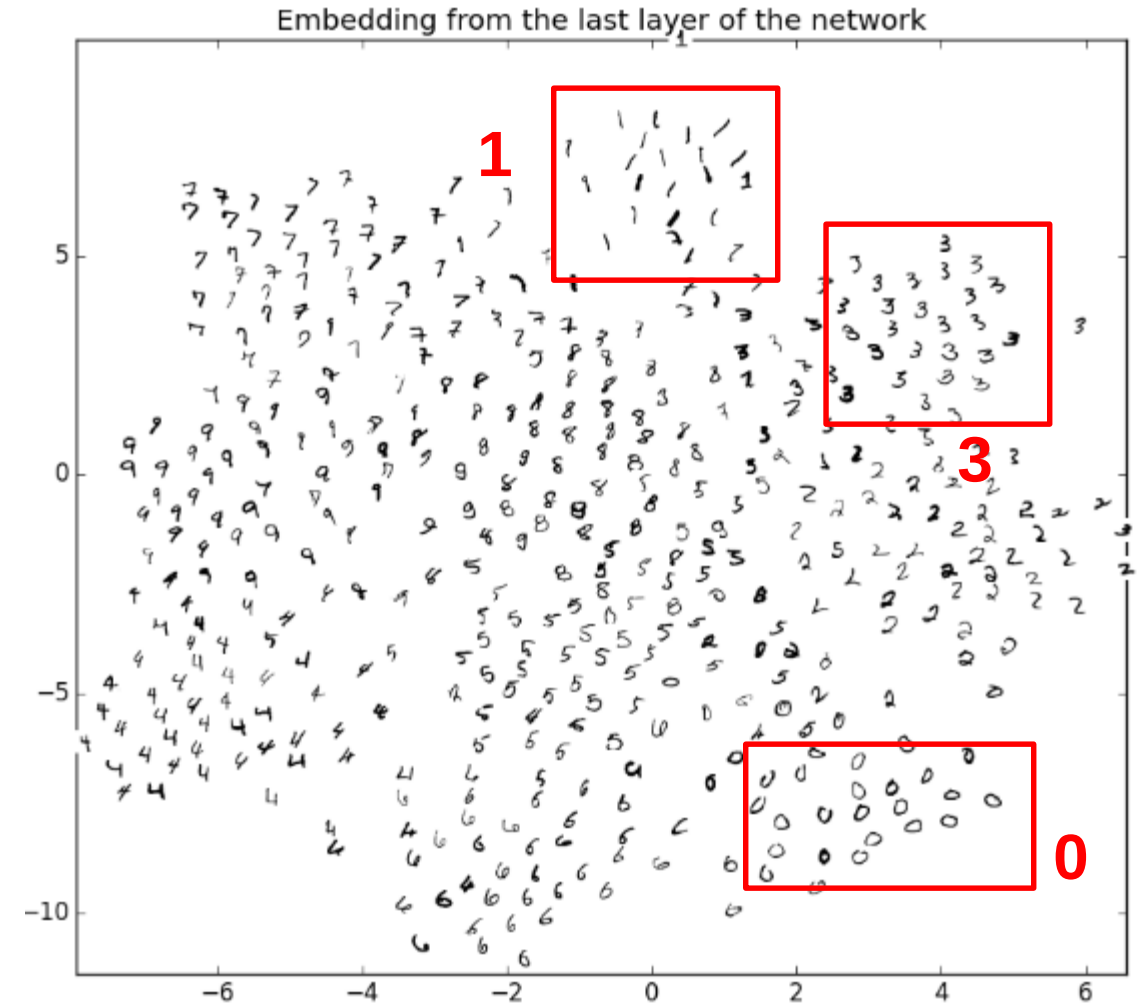
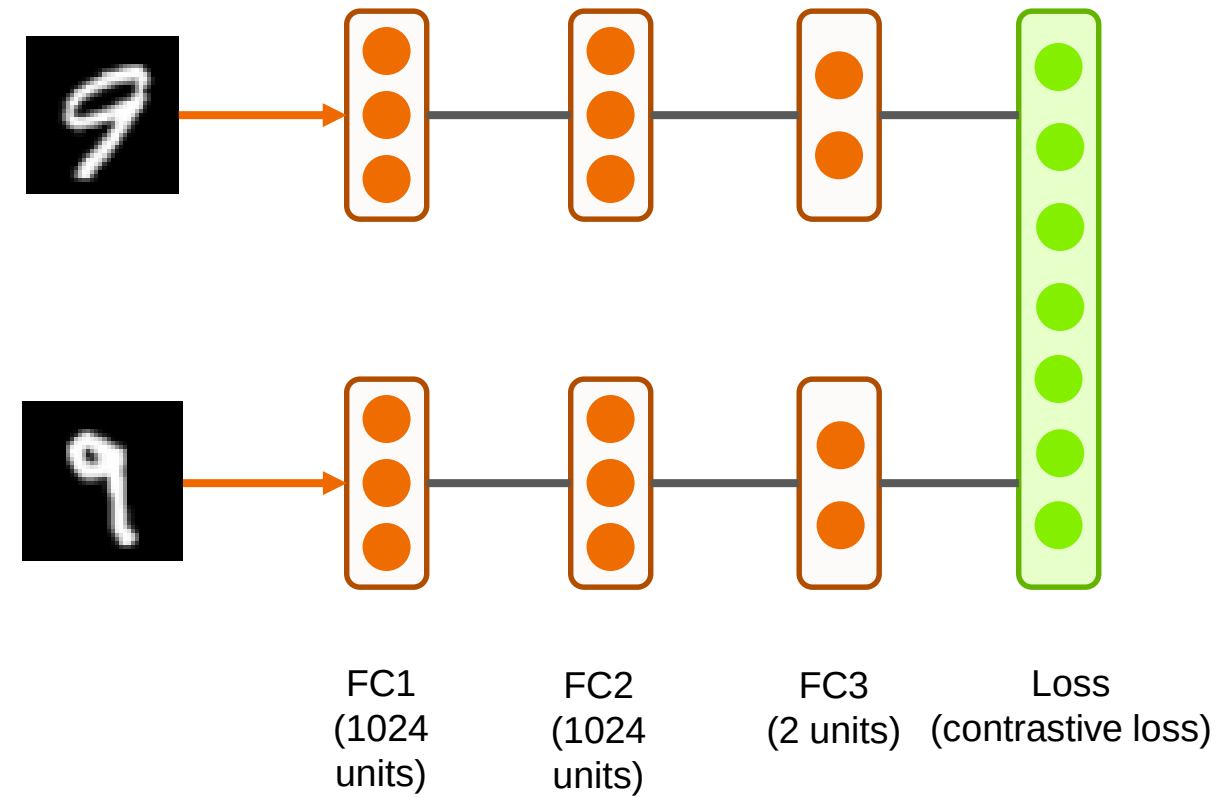
Similar images (neighbors  
in the neighborhood graph)

Make this large

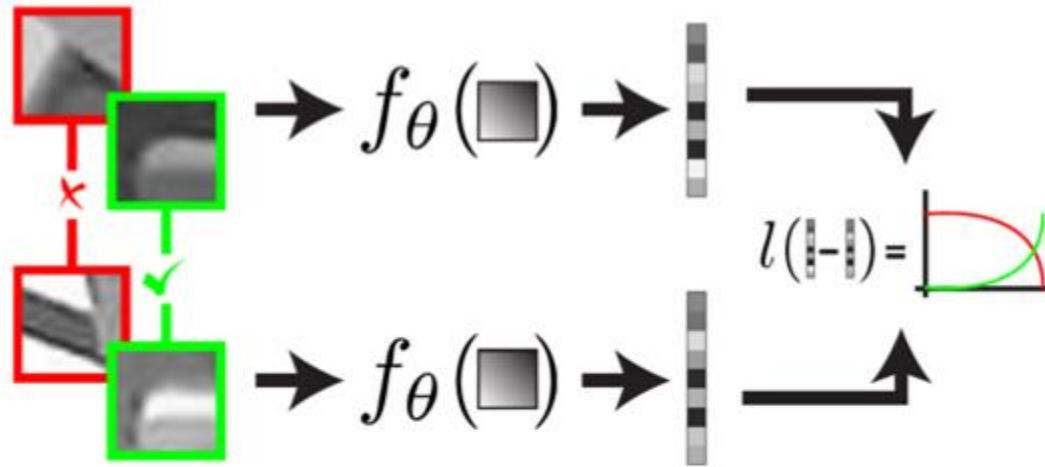


Dissimilar images  
(non-neighbors in the  
neighborhood graph)

# Example: MNIST

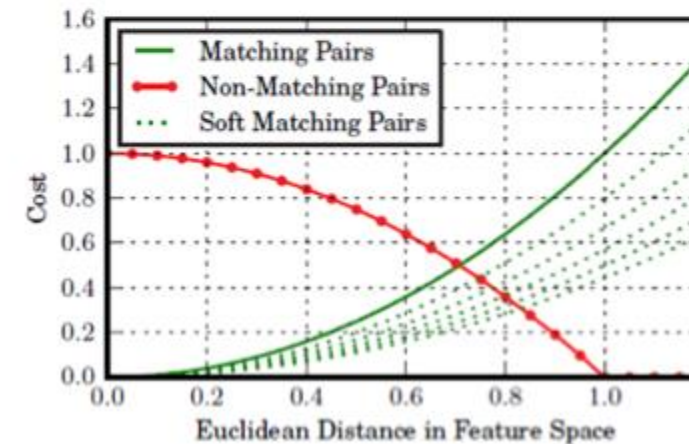


# Application: Learning to Match Siamese Network

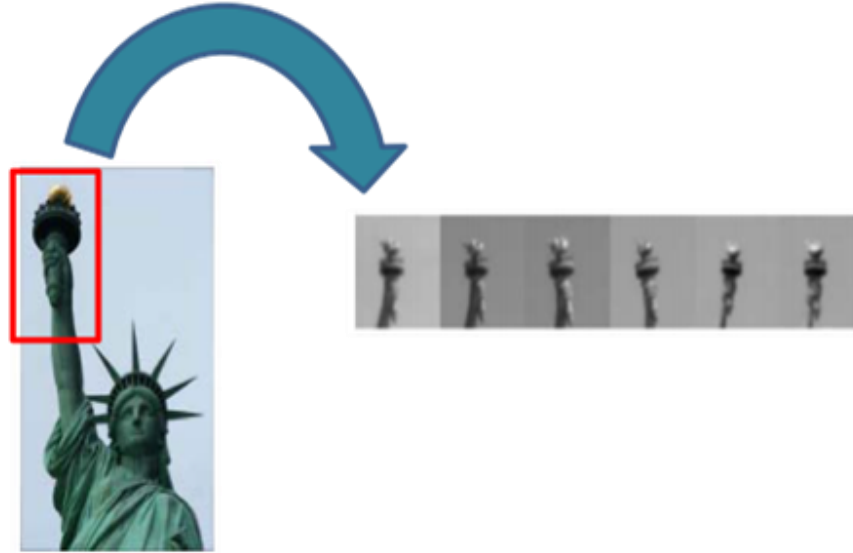


Using the contrastive cost function

$$l_\theta(y_i, y_j) = \begin{cases} s_{ij} d_{ij}^2, & \text{if matching} \\ \max(1.0 - d_{ij}^2, 0), & \text{if non-matching} \end{cases}$$



# Matching across views



Learn a discriminative representation of patches from different views of 3D points

Smo-Serra, E, Trulls, E, Ferraz, L, Kokkinos, I., Fua, P. and Moreno-Noguer, F., 2015. Discriminative learning of deep convolutional feature point descriptors. In *Proceedings of the IEEE International Conference on Computer Vision* (pp. 118-126).

# Application

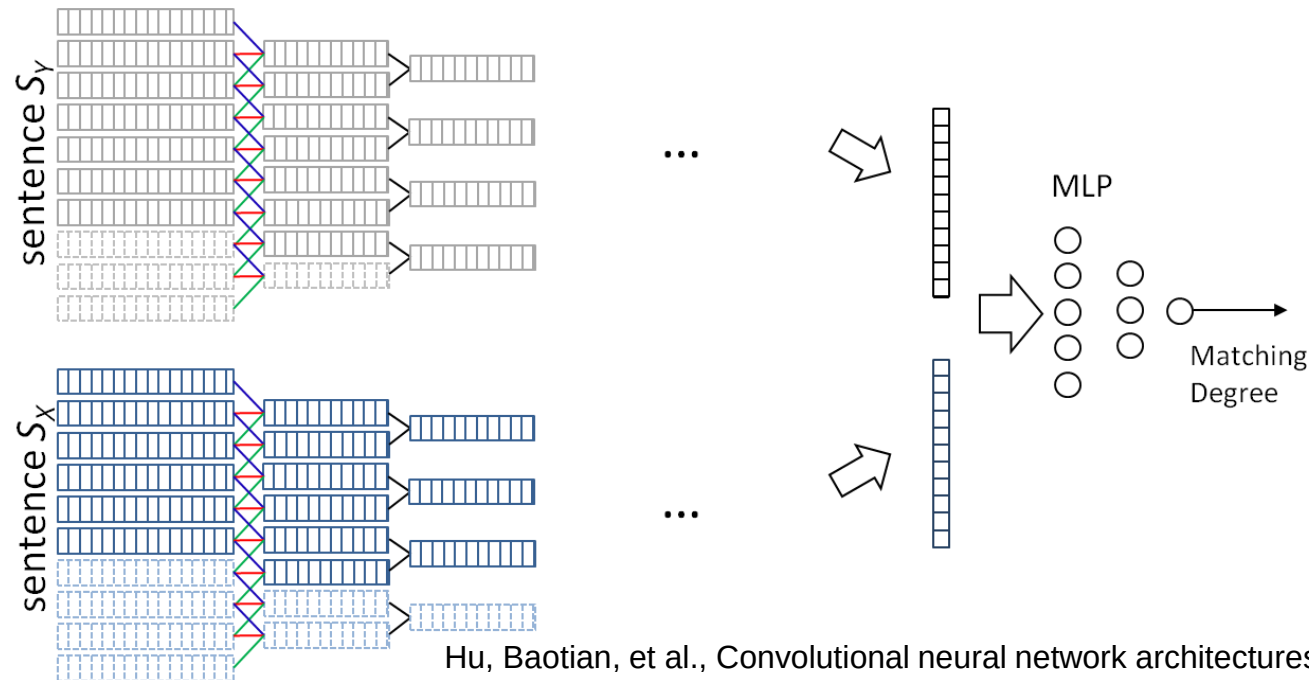
- **Application:** Sentence completion, response to tweet, paraphrase identification

## Example:

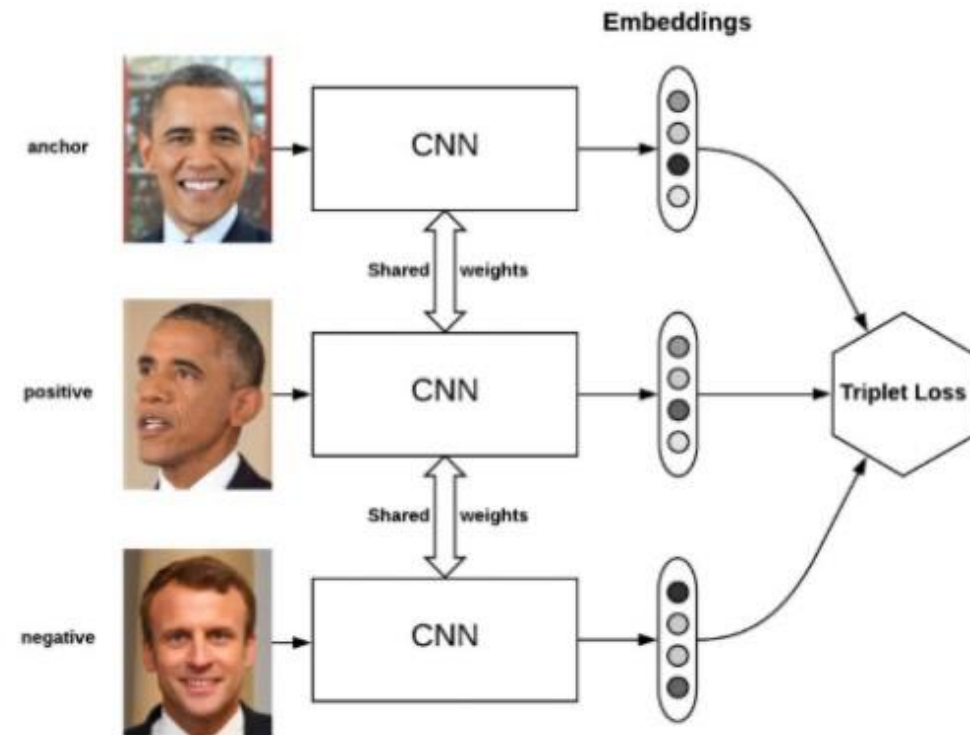
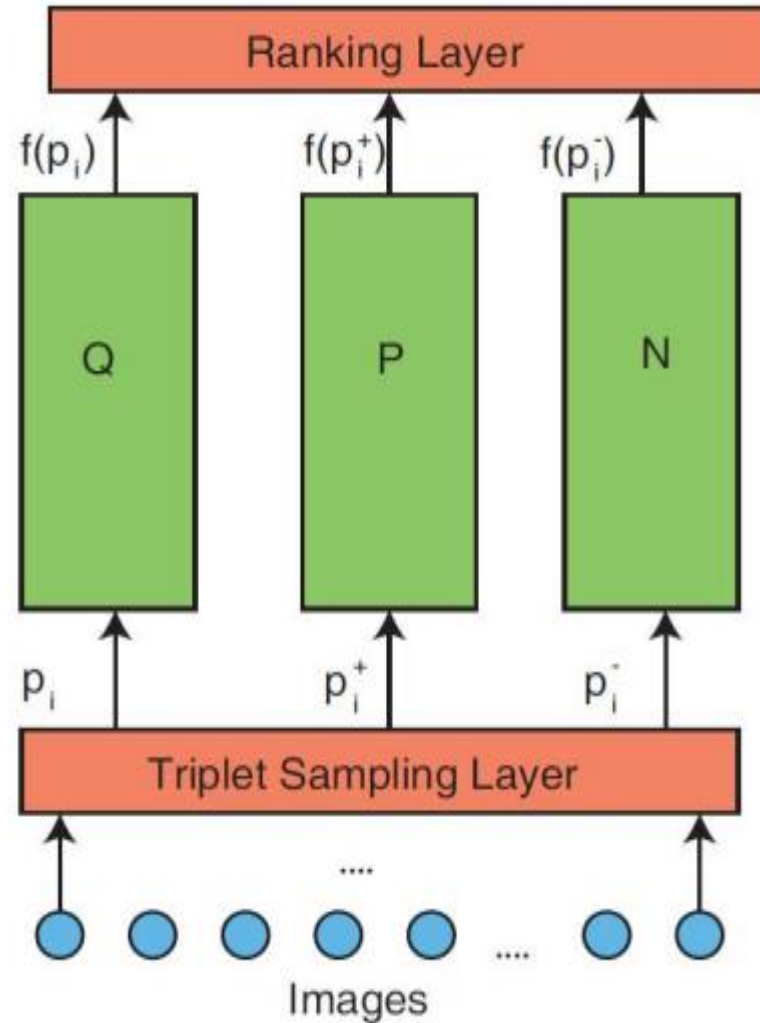
word2vec



- x : Damn, I have to work overtime this weekend!
- Y<sup>+</sup>: Try to have some rest buddy.
- y: It is hard to find a job, better start polishing your resume.

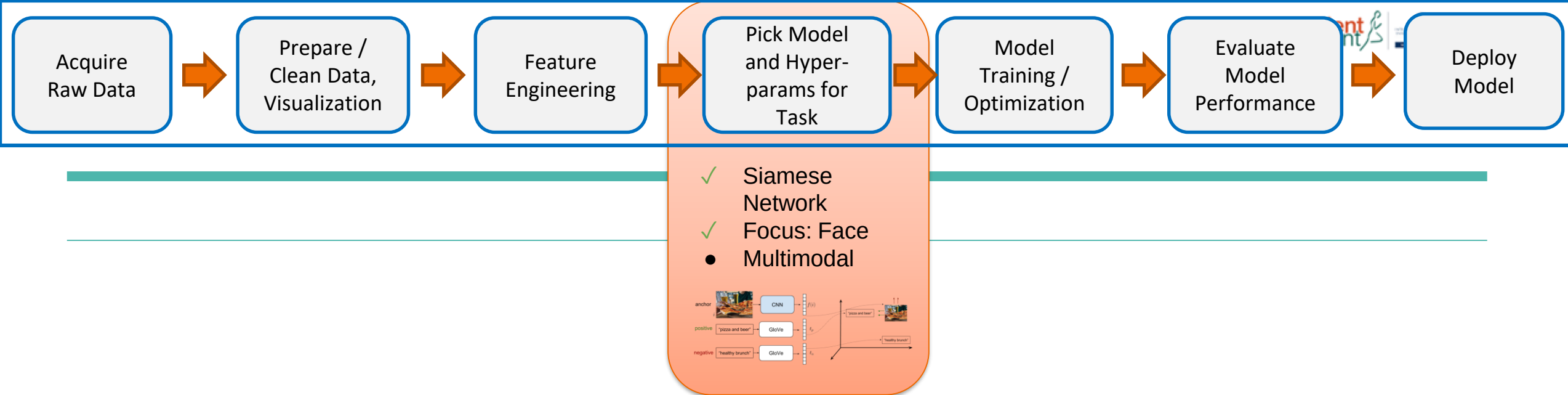


# Triplet Loss



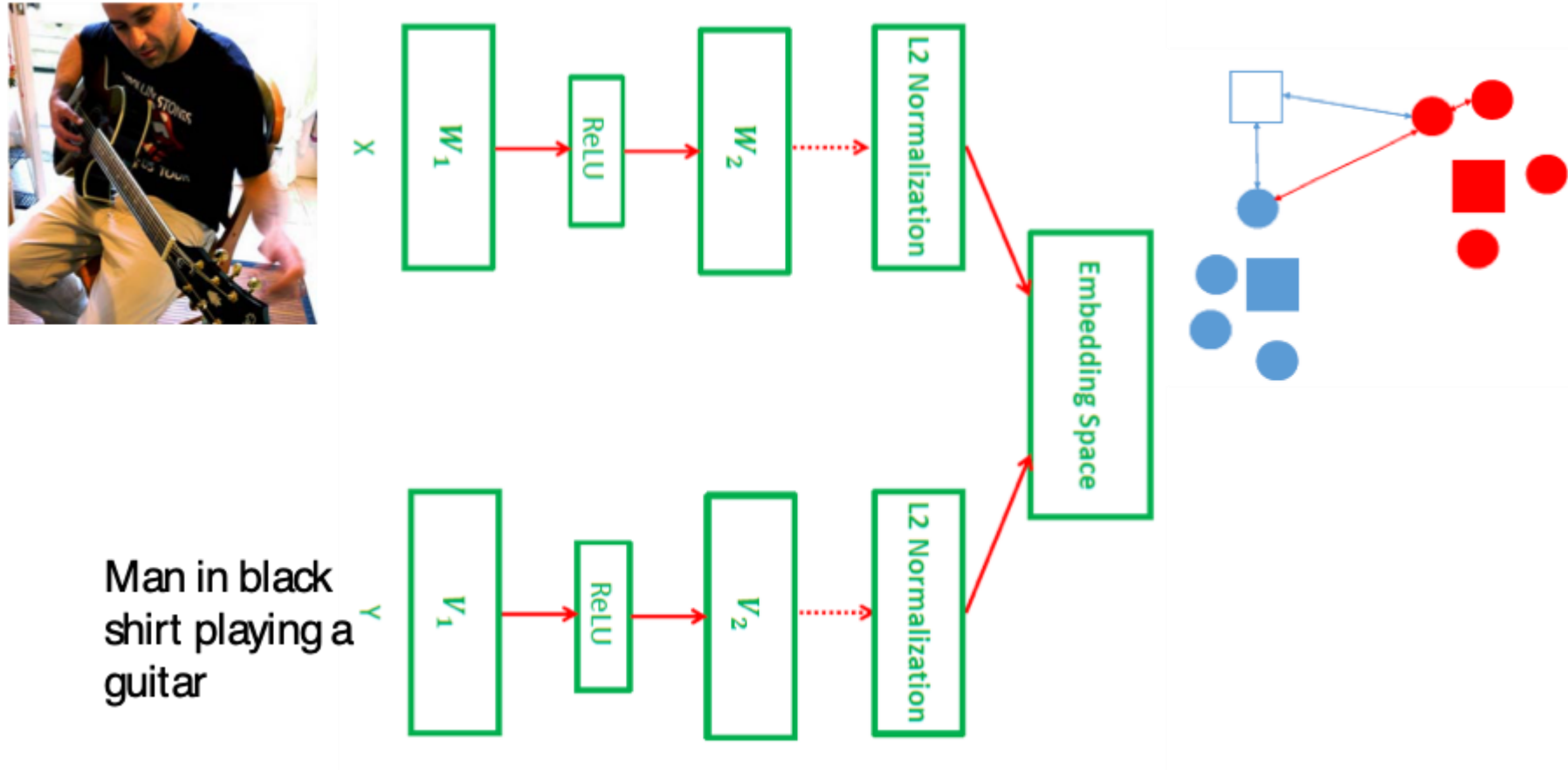
# Image Retrieval (Ranking)

|          |                                                                                    |                                                                                      |                                                                                      |                                                                                      |                                                                                      |
|----------|------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| Query    |   |    |   |   |   |
| Positive |   |    |   |   |   |
| Negative |  |  |  |  |  |



## Extension: Multimodal

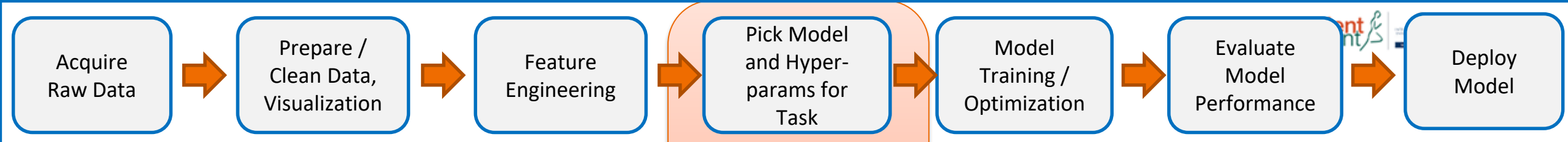
# Cross modal matching



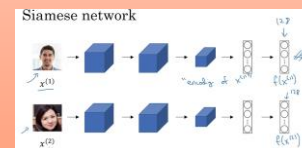
Wang, L., Li, Y. and Lazebnik, S., 2016. Learning deep structure-preserving image-text embeddings. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition* (pp. 5005-5013).

# Summary

- Siamese and Triplet networks/losses are popular for solving fine grain classification (e.g. Face), capturing subjective needs (eg. Invariant to rotation), rankings etc.
- Available in many frameworks.
- Number of examples increase (e.g.  ${}^nC_2$ ,  ${}^nC_3$ ). Finding right pairs/triplets is also important.



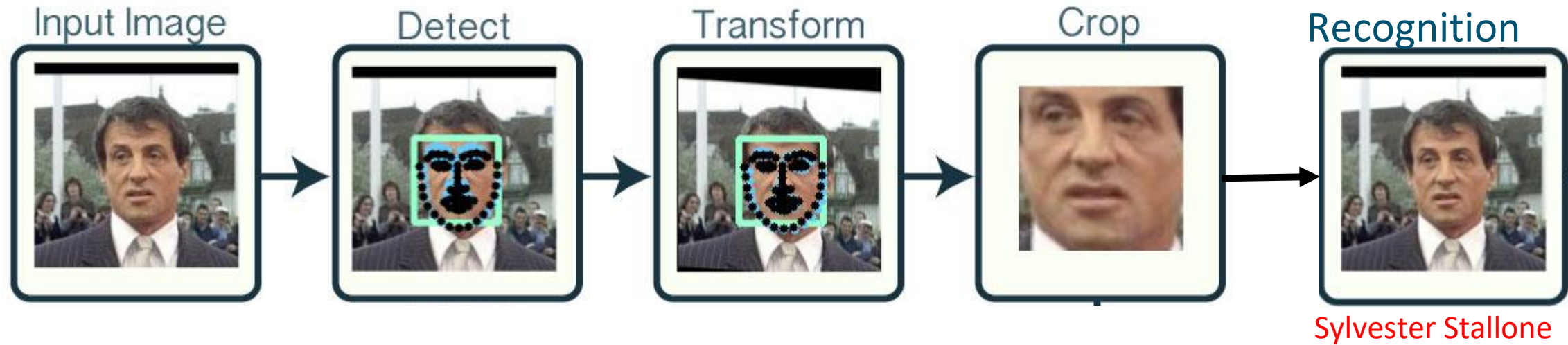
- ✓ Siamese Network
  - Focus: Face



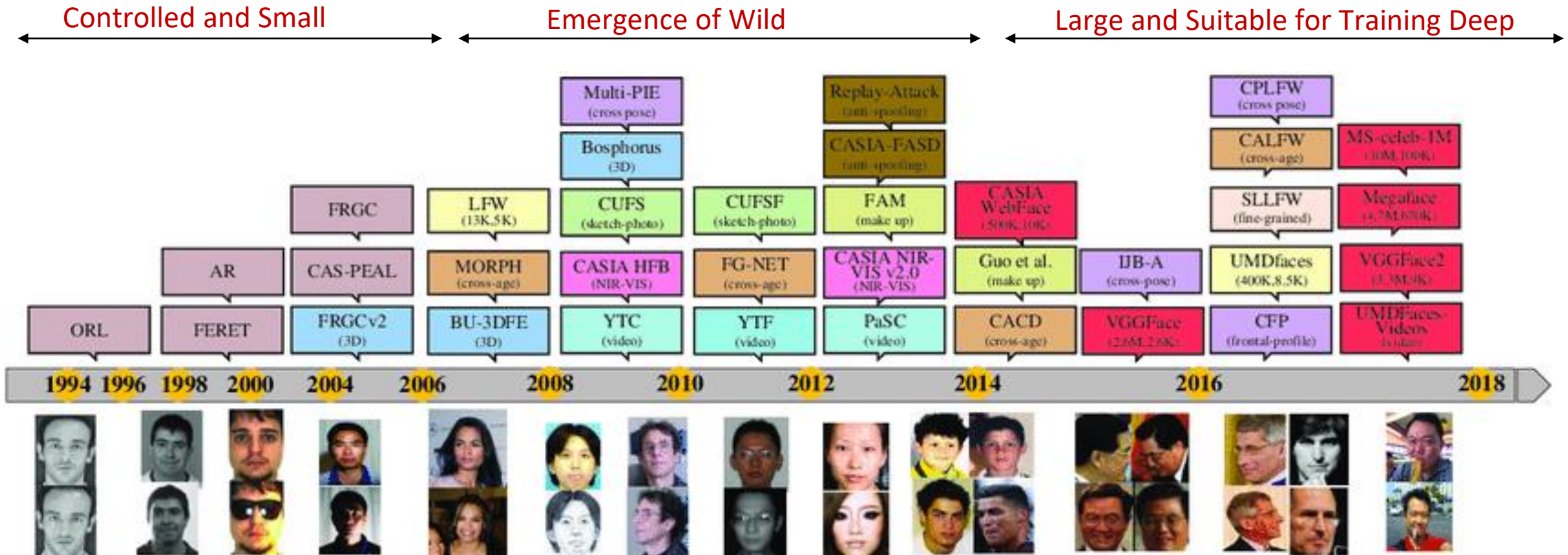
## Example: Face

Focus on a specific area to appreciate the developments

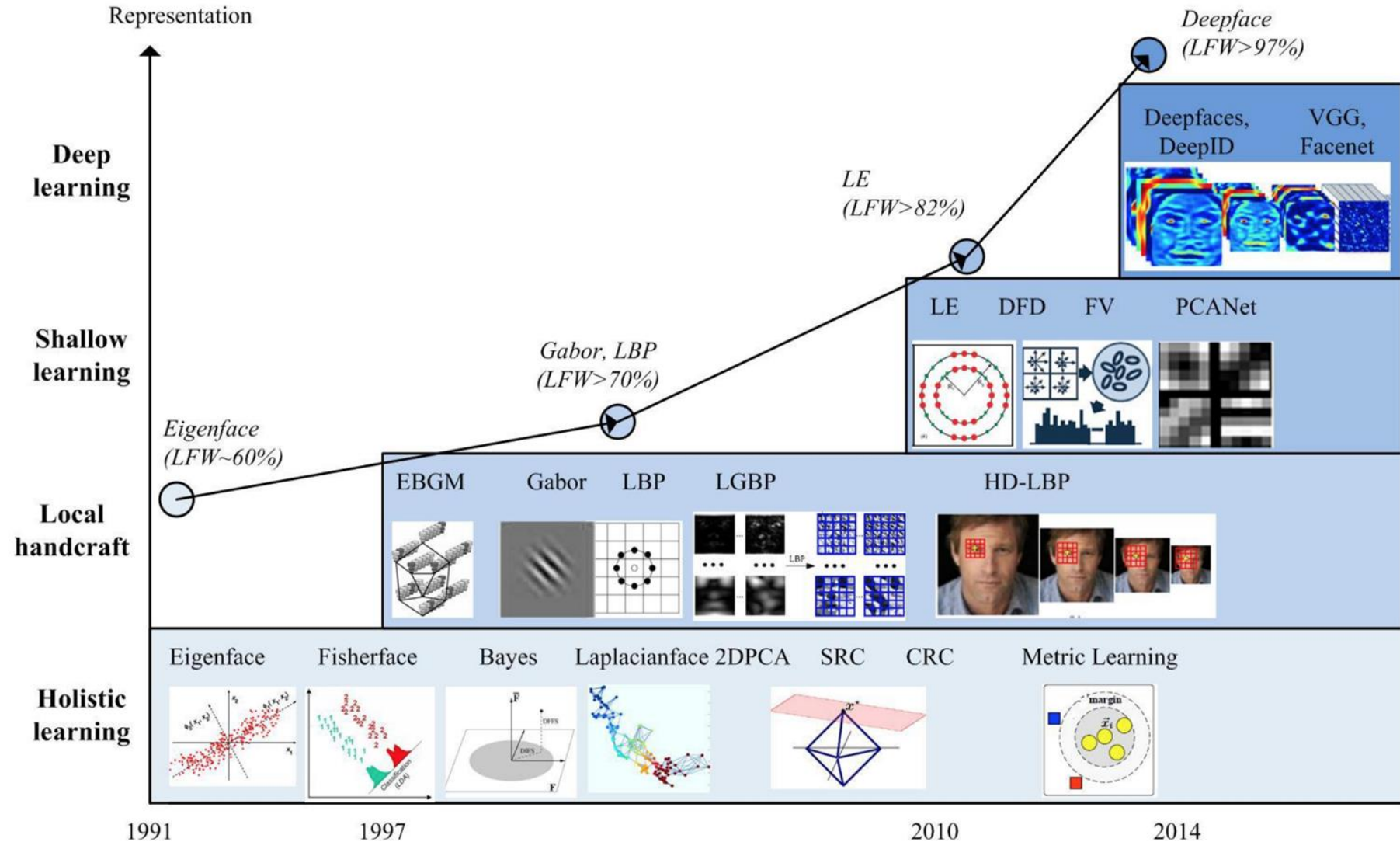
# Face processing pipeline



# Evolution of FR Datasets



# Performance on LFW

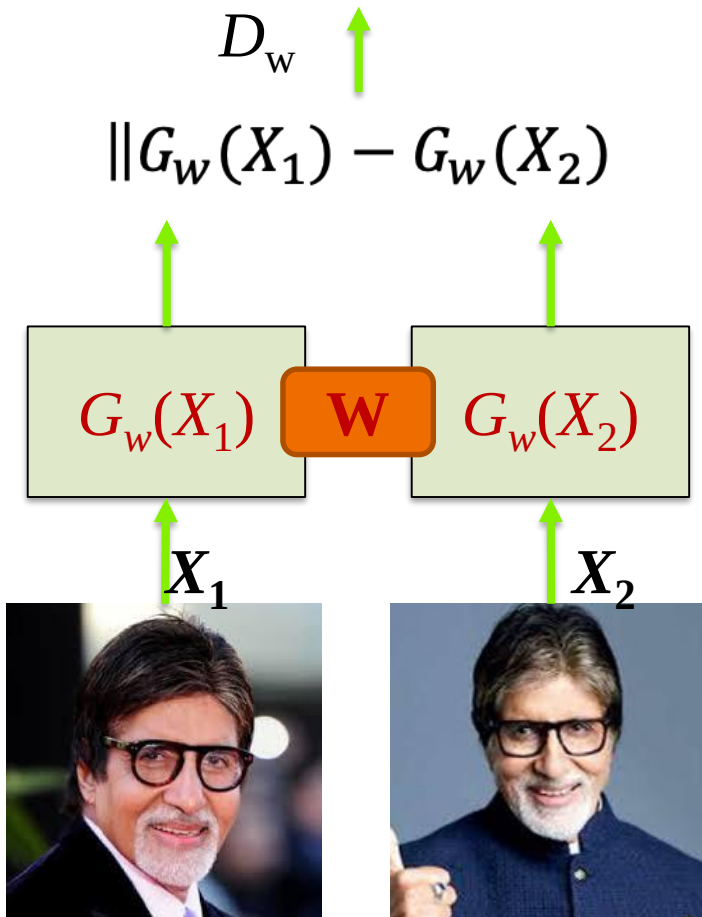


# Systematic Evolution of Performance (LFW)

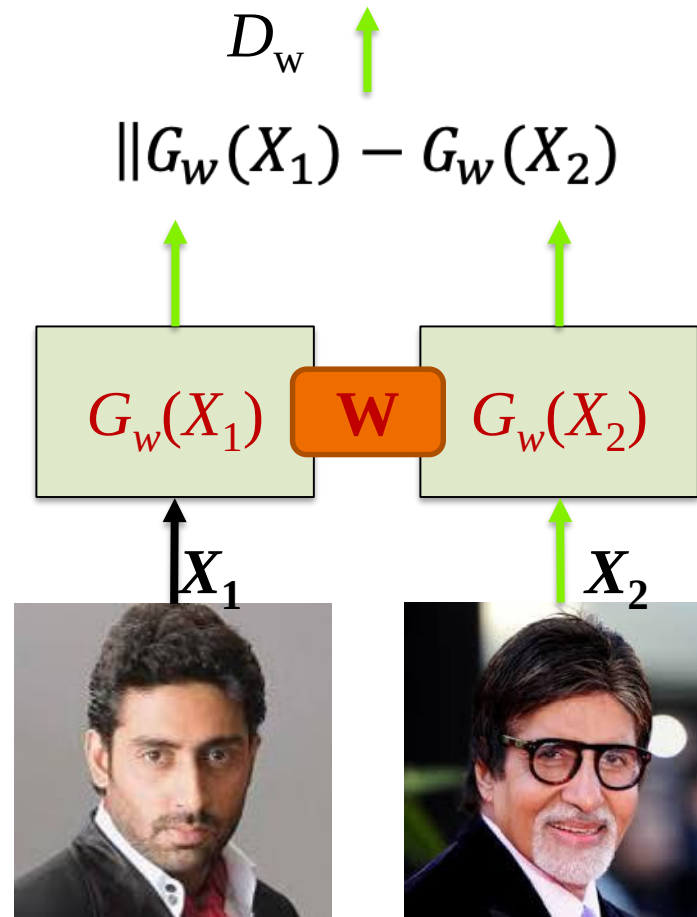
| Method              | Public. Time | Loss                  | Architecture | Number of Networks | Training Set                                   | Accuracy $\pm$ Std(%) |
|---------------------|--------------|-----------------------|--------------|--------------------|------------------------------------------------|-----------------------|
| DeepFace [20]       | 2014         | softmax               | Alexnet      | 3                  | Facebook (4.4M,4K)                             | 97.35 $\pm$ 0.25      |
| DeepID2 [21]        | 2014         | contrastive loss      | Alexnet      | 25                 | CelebFaces+ (0.2M,10K)                         | 99.15 $\pm$ 0.13      |
| DeepID3 [36]        | 2015         | contrastive loss      | VGGNet-10    | 50                 | CelebFaces+ (0.2M,10K)                         | 99.53 $\pm$ 0.10      |
| FaceNet [38]        | 2015         | triplet loss          | GoogleNet-24 | 1                  | Google (500M,10M)                              | 99.63 $\pm$ 0.09      |
| Baidu [58]          | 2015         | triplet loss          | CNN-9        | 10                 | Baidu (1.2M,18K)                               | 99.77                 |
| VGGface [37]        | 2015         | triplet loss          | VGGNet-16    | 1                  | VGGface (2.6M,2.6K)                            | 98.95                 |
| light-CNN [85]      | 2015         | softmax               | light CNN    | 1                  | MS-Celeb-1M (8.4M,100K)                        | 98.8                  |
| Center Loss [101]   | 2016         | center loss           | Lenet+-7     | 1                  | CASIA-WebFace, CACD2000, Celebrity+ (0.7M,17K) | 99.28                 |
| L-softmax [104]     | 2016         | L-softmax             | VGGNet-18    | 1                  | CASIA-WebFace (0.49M,10K)                      | 98.71                 |
| Range Loss [82]     | 2016         | range loss            | VGGNet-16    | 1                  | MS-Celeb-1M, CASIA-WebFace (5M,100K)           | 99.52                 |
| L2-softmax [109]    | 2017         | L2-softmax            | ResNet-101   | 1                  | MS-Celeb-1M (3.7M,58K)                         | 99.78                 |
| Normface [110]      | 2017         | contrastive loss      | ResNet-28    | 1                  | CASIA-WebFace (0.49M,10K)                      | 99.19                 |
| CoCo loss [112]     | 2017         | CoCo loss             | -            | 1                  | MS-Celeb-1M (3M,80K)                           | 99.86                 |
| vMF loss [115]      | 2017         | vMF loss              | ResNet-27    | 1                  | MS-Celeb-1M (4.6M,60K)                         | 99.58                 |
| Marginal Loss [116] | 2017         | marginal loss         | ResNet-27    | 1                  | MS-Celeb-1M (4M,80K)                           | 99.48                 |
| SphereFace [84]     | 2017         | A-softmax             | ResNet-64    | 1                  | CASIA-WebFace (0.49M,10K)                      | 99.42                 |
| CCL [113]           | 2018         | center invariant loss | ResNet-27    | 1                  | CASIA-WebFace (0.49M,10K)                      | 99.12                 |
| AMS loss [105]      | 2018         | AMS loss              | ResNet-20    | 1                  | CASIA-WebFace (0.49M,10K)                      | 99.12                 |
| Cosface [107]       | 2018         | cosface               | ResNet-64    | 1                  | CASIA-WebFace (0.49M,10K)                      | 99.33                 |
| Arcface [106]       | 2018         | arcface               | ResNet-100   | 1                  | MS-Celeb-1M (3.8M,85K)                         | 99.83                 |
| Ring loss [117]     | 2018         | Ring loss             | ResNet-64    | 1                  | MS-Celeb-1M (3.5M,31K)                         | 99.50                 |

# Siamese Loss

Make this smaller

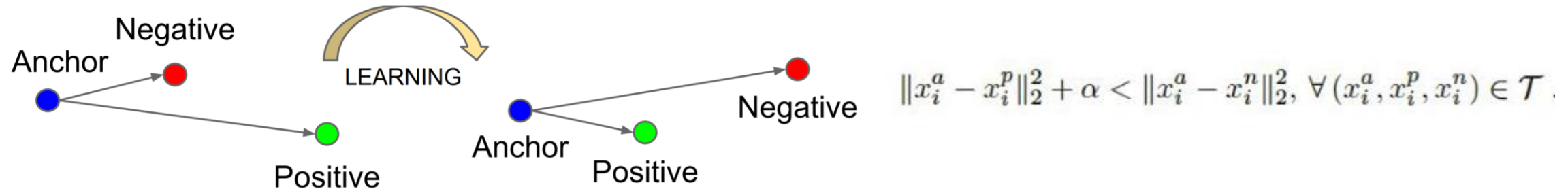


Make this larger



- Only pair-wise Labels
- Similarity Metric:  
 $D_w(X_1, X_2)$
- Have shared weights
- Training in batches

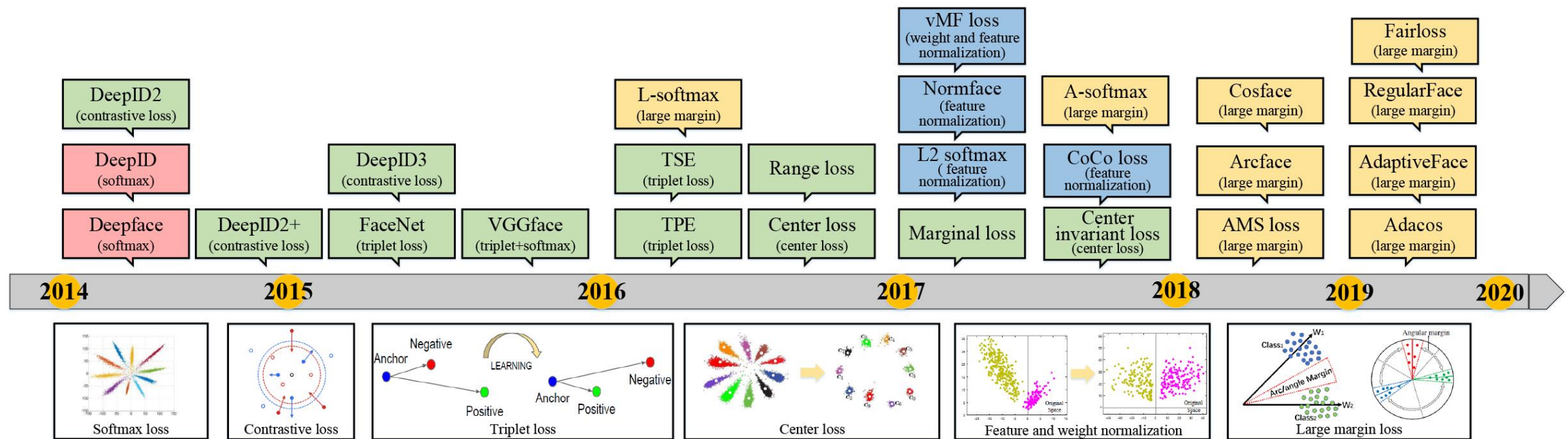
# Triplet loss function



where  $f$  is the embedding

$$\sum_i^N \left[ \|f(x_i^a) - f(x_i^p)\|_2^2 - \|f(x_i^a) - f(x_i^n)\|_2^2 + \alpha \right]_+$$

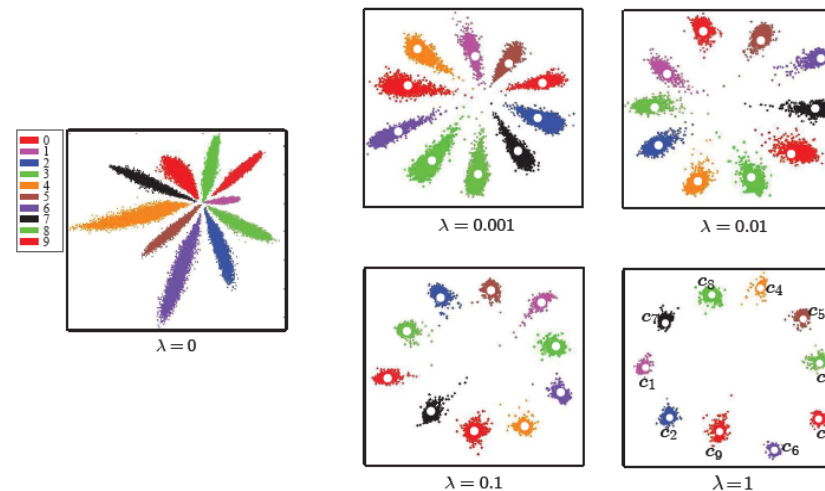
# Summary: Evolution of Loss Functions



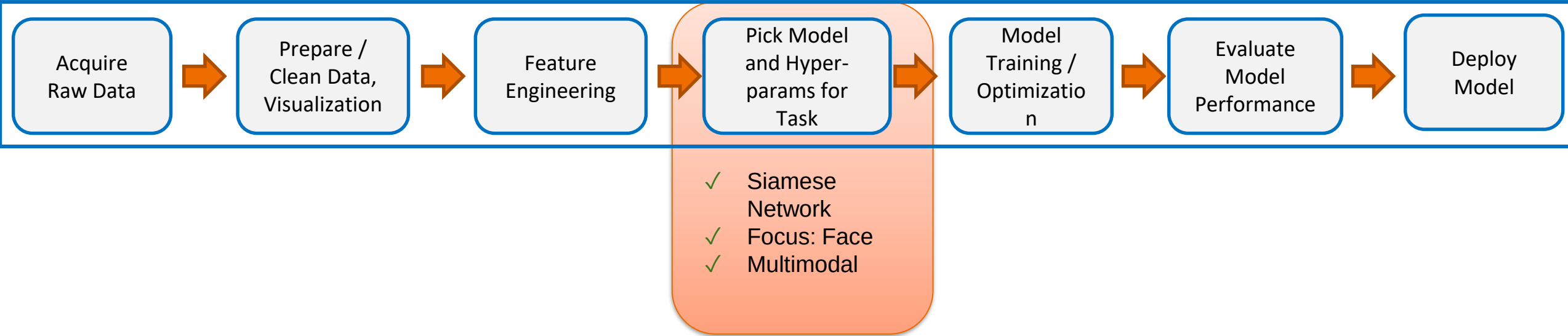
# Center Loss

$$L = L_s + \mu L_c = -\sum_{i=1}^m \log \frac{e^{W_{y_i}^l x_i + b_{y_i}}}{\sum_{j=1}^n e^{W_j^T x_i + b_j}} + \frac{\lambda}{2} \sum_{i=1}^m ||x_i - c_{y_j}||_2^2$$

Two Terms (1) For Interclass separability (softmax) and (2) Intraclass compactness and Lambda that balance between them



# Summary



**Thanks!!**

**Questions?**